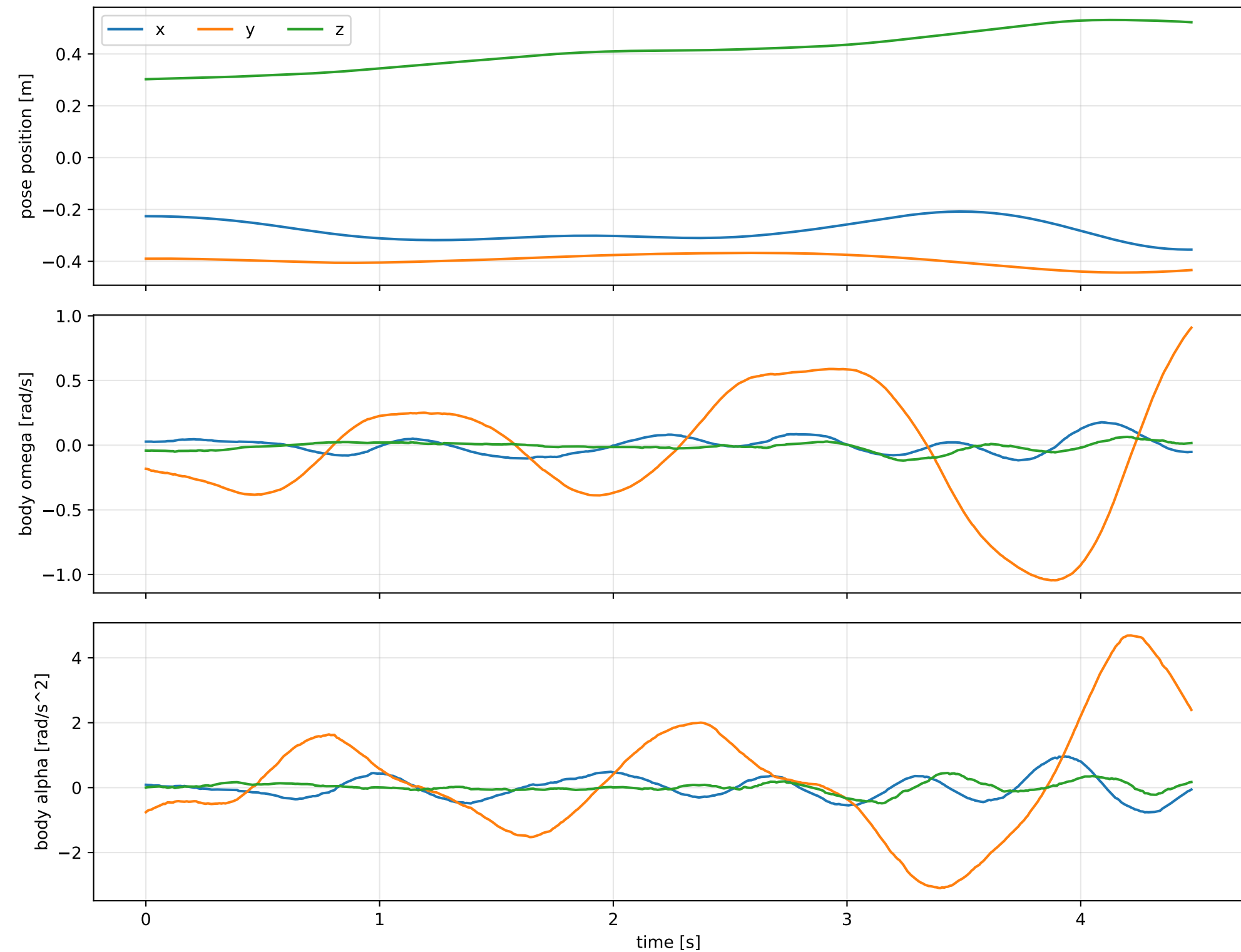
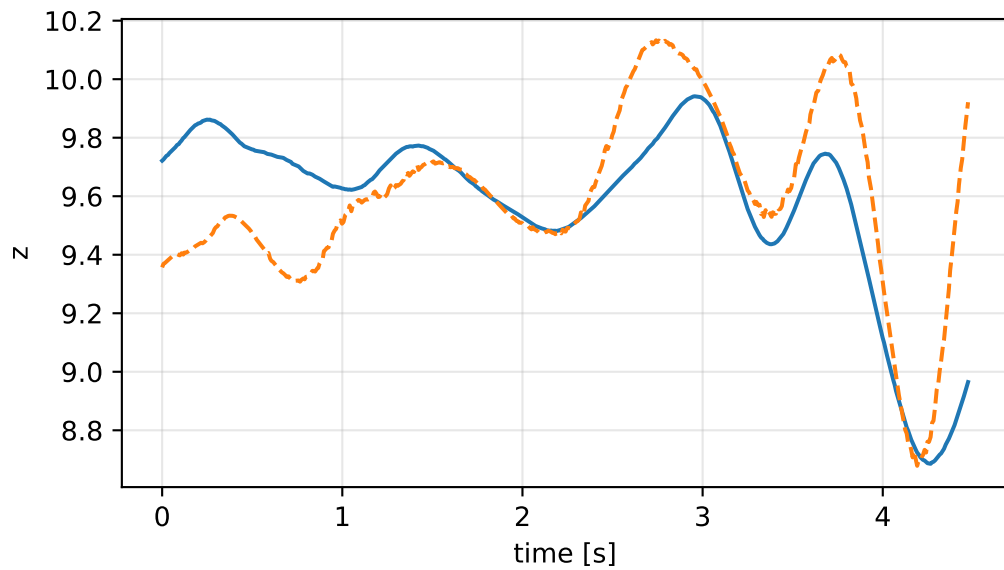
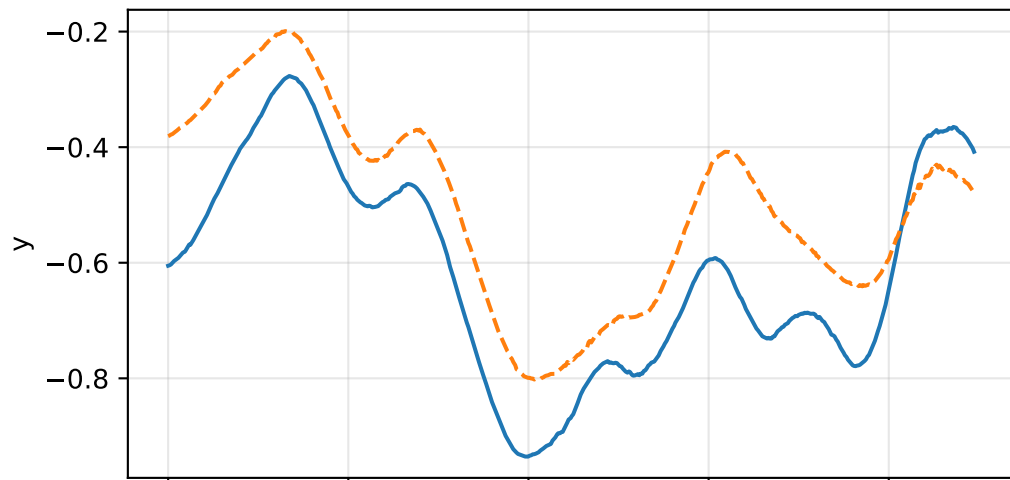
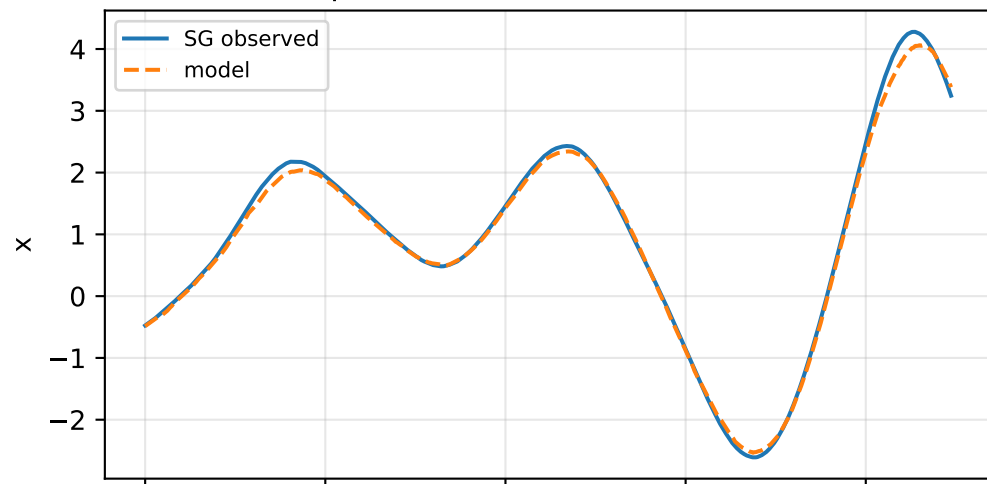


so3_naive_rotation_vector_derivatives: SG trajectory and derived motion

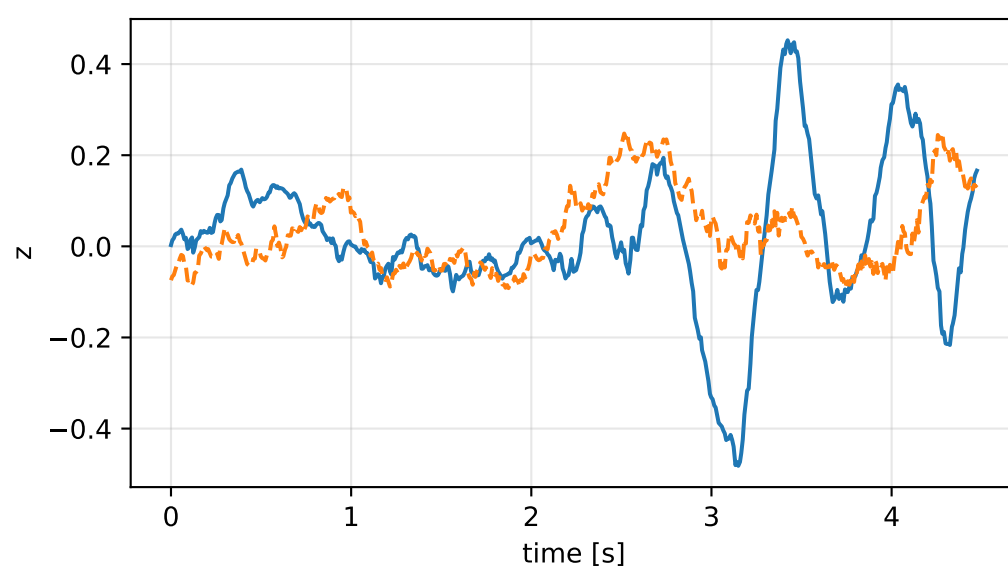
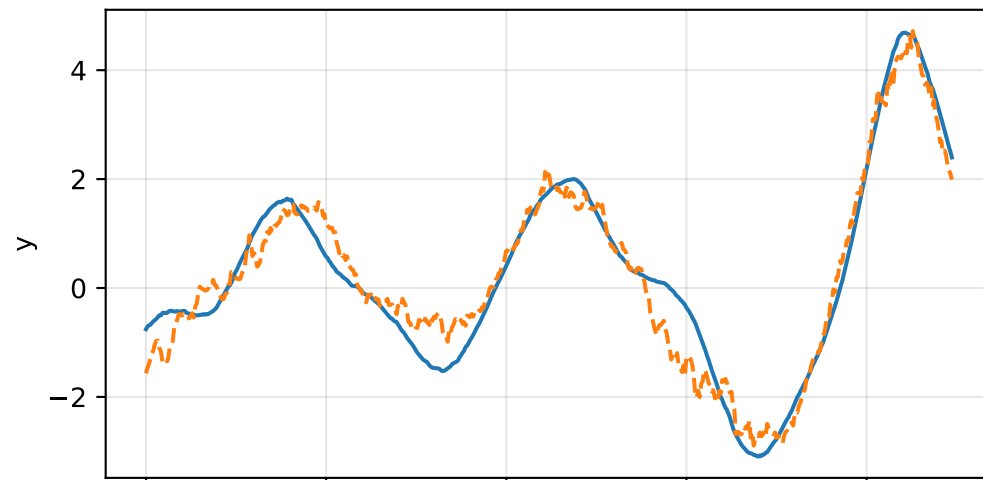
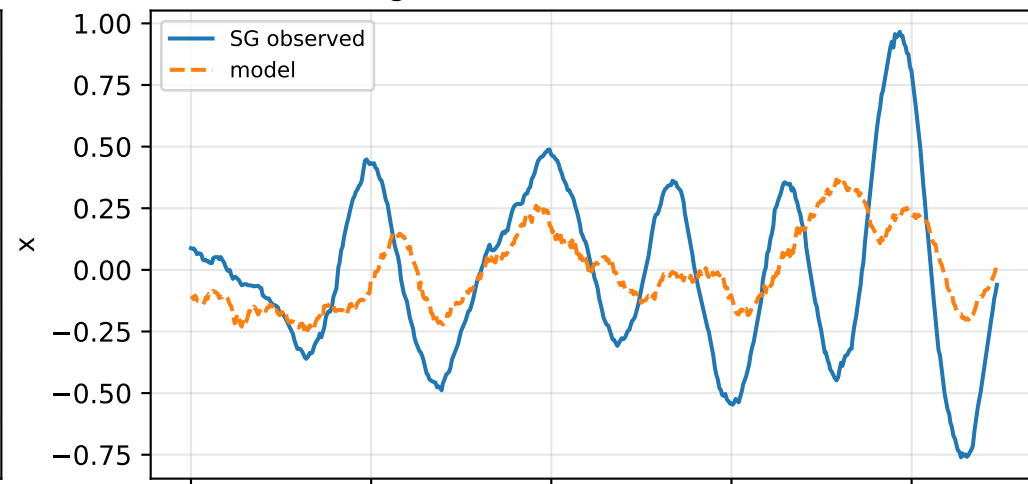


so3_naive_rotation_vector_derivatives: acceleration residual objective

specific acceleration [m/s^2]

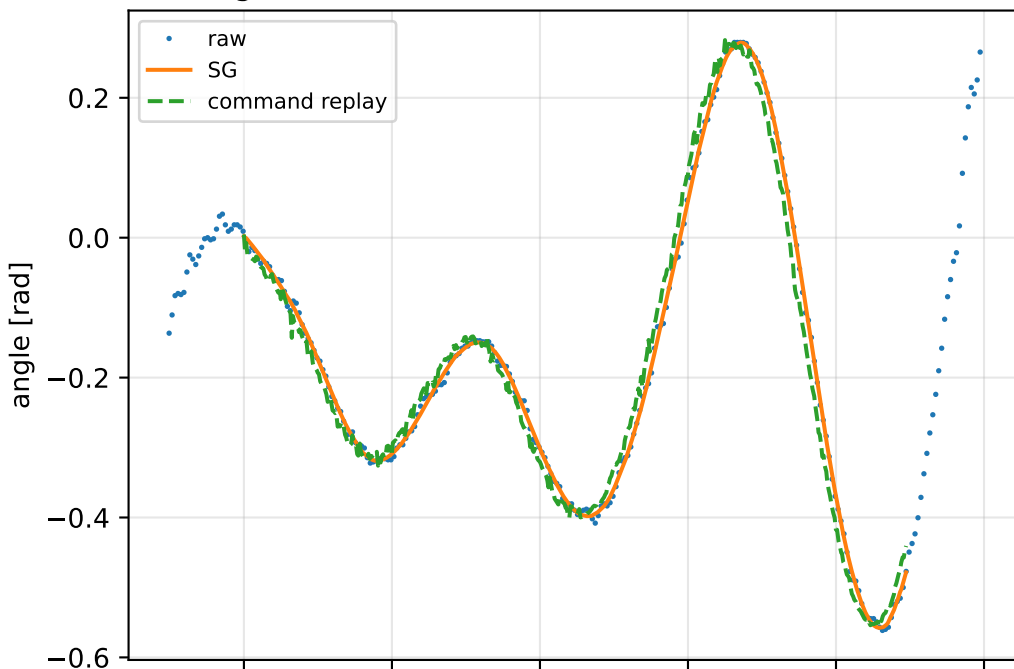


angular acceleration [rad/s^2]

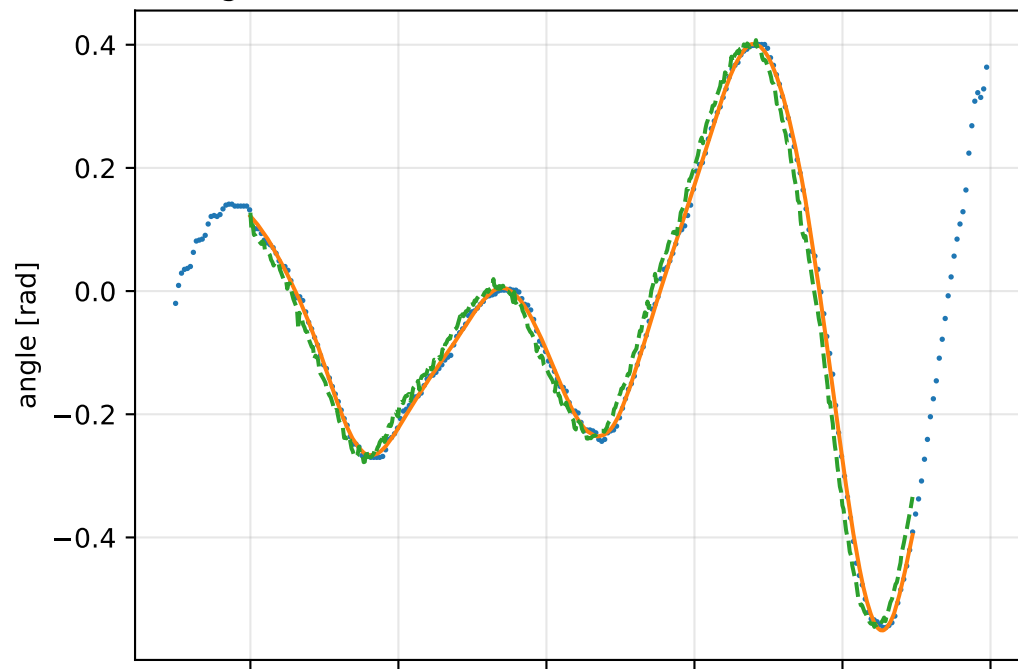


so3_naive_rotation_vector_derivatives: gimbal measurement audit (objective=measured_sg)

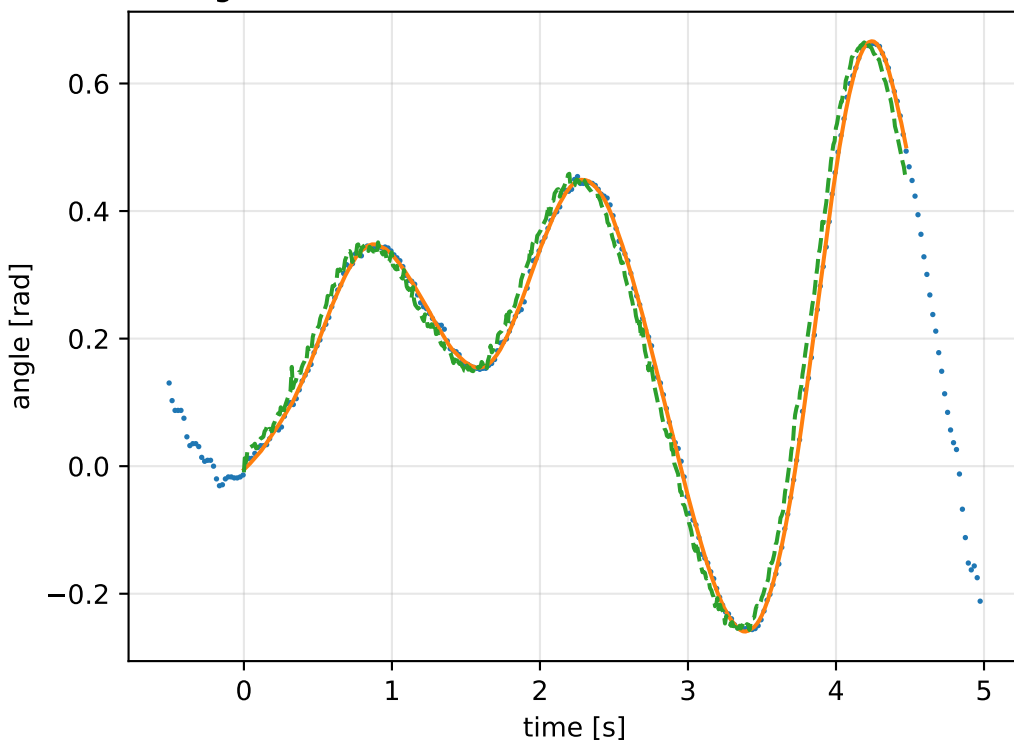
gimbal 1: RMSE=0.027, max=0.07529 rad



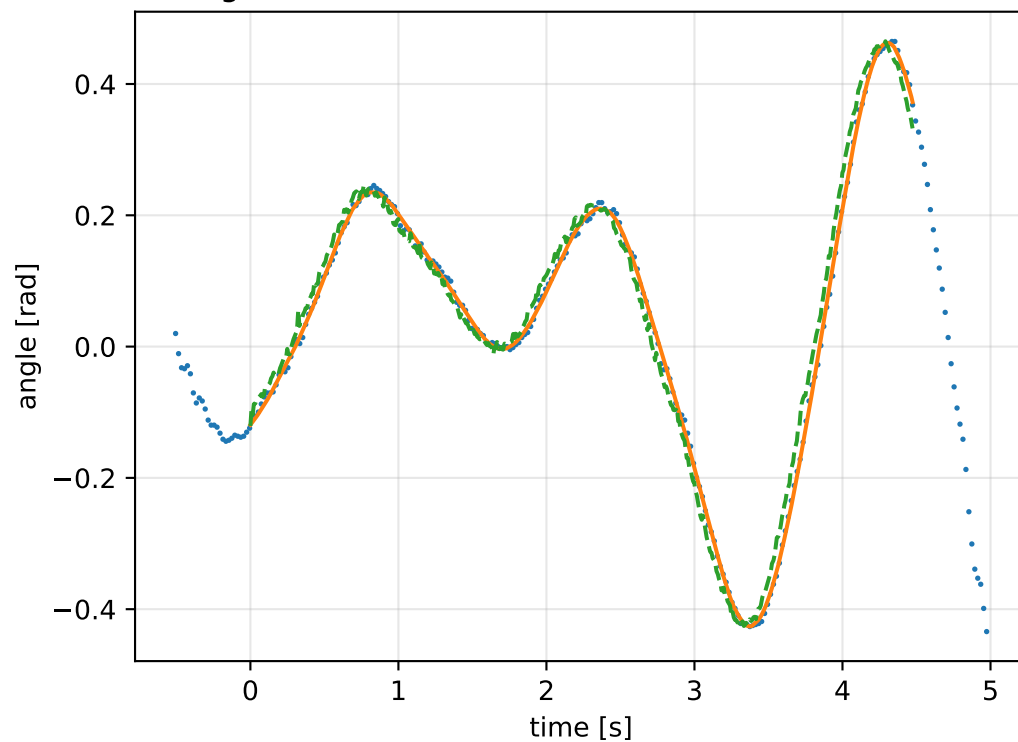
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

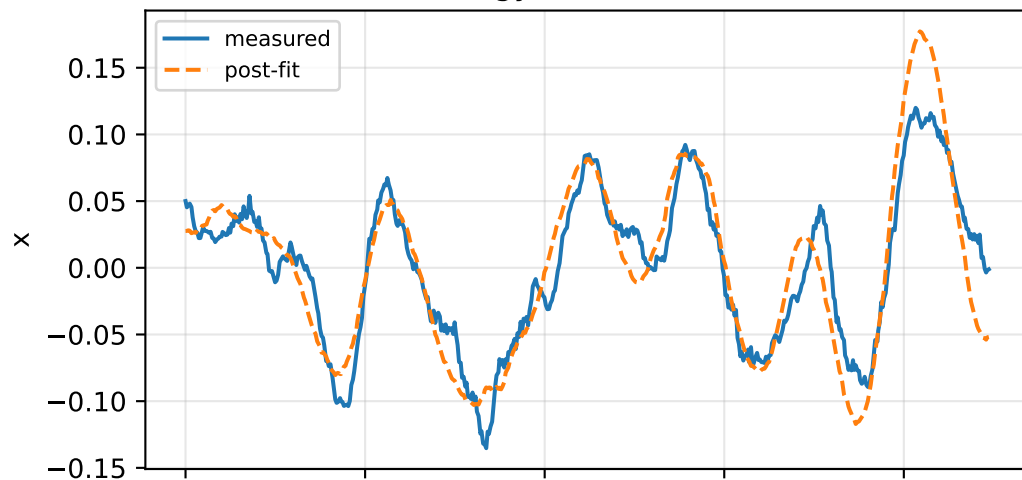


gimbal 4: RMSE=0.02598, max=0.07261 rad

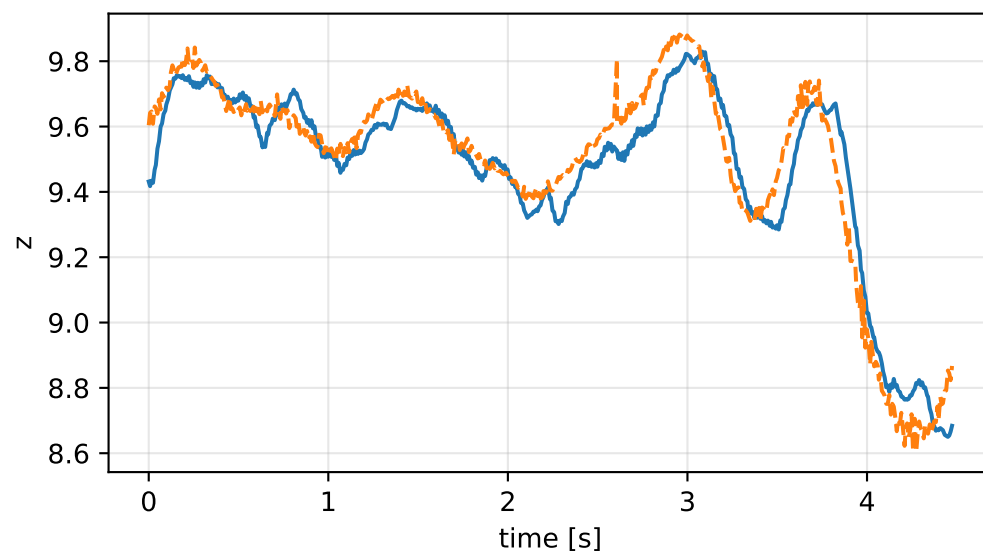
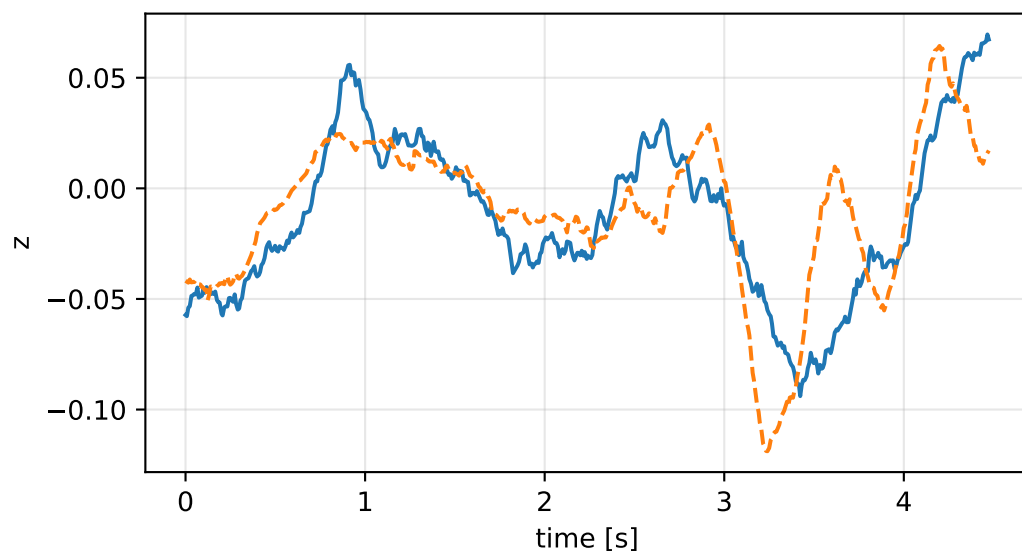
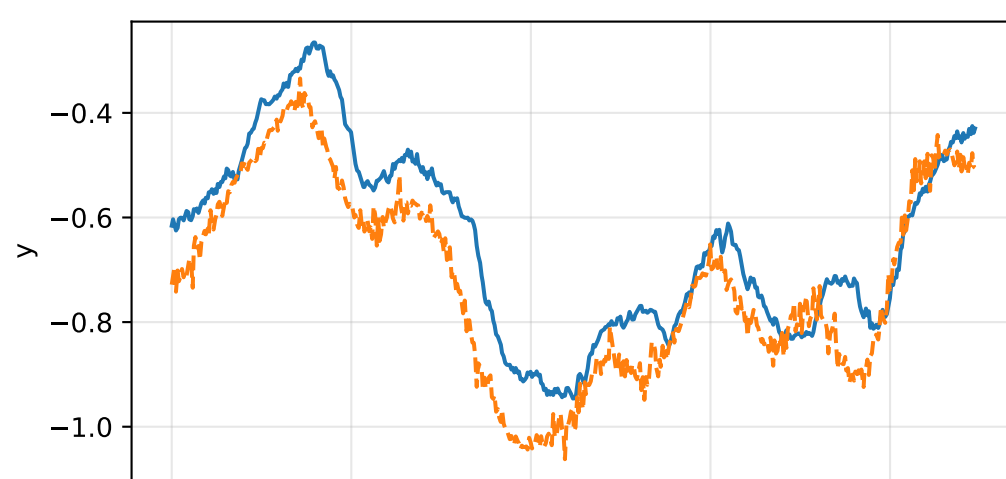
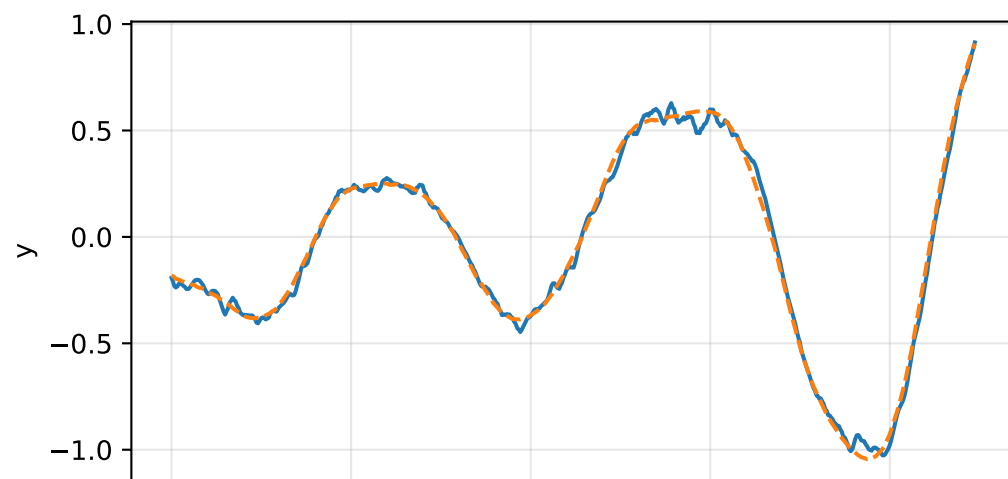
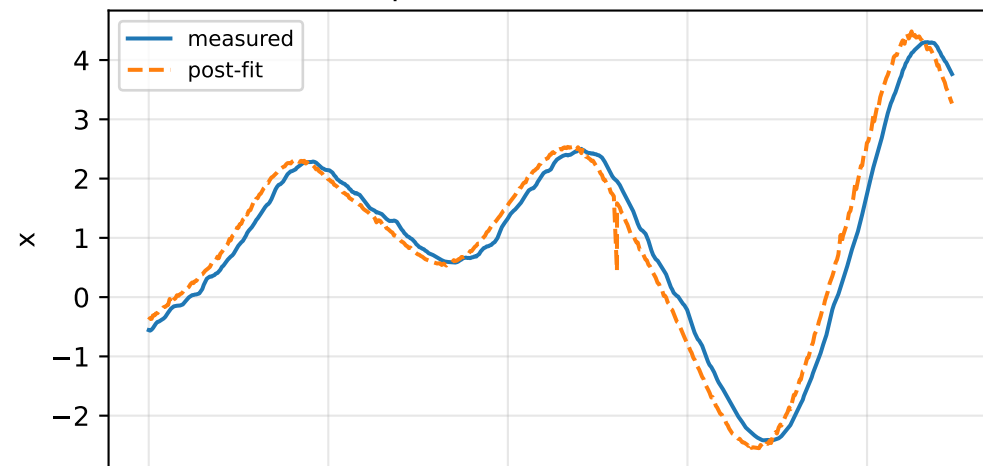


so3_naive_rotation_vector_derivatives: IMU comparison (diagnostic only)

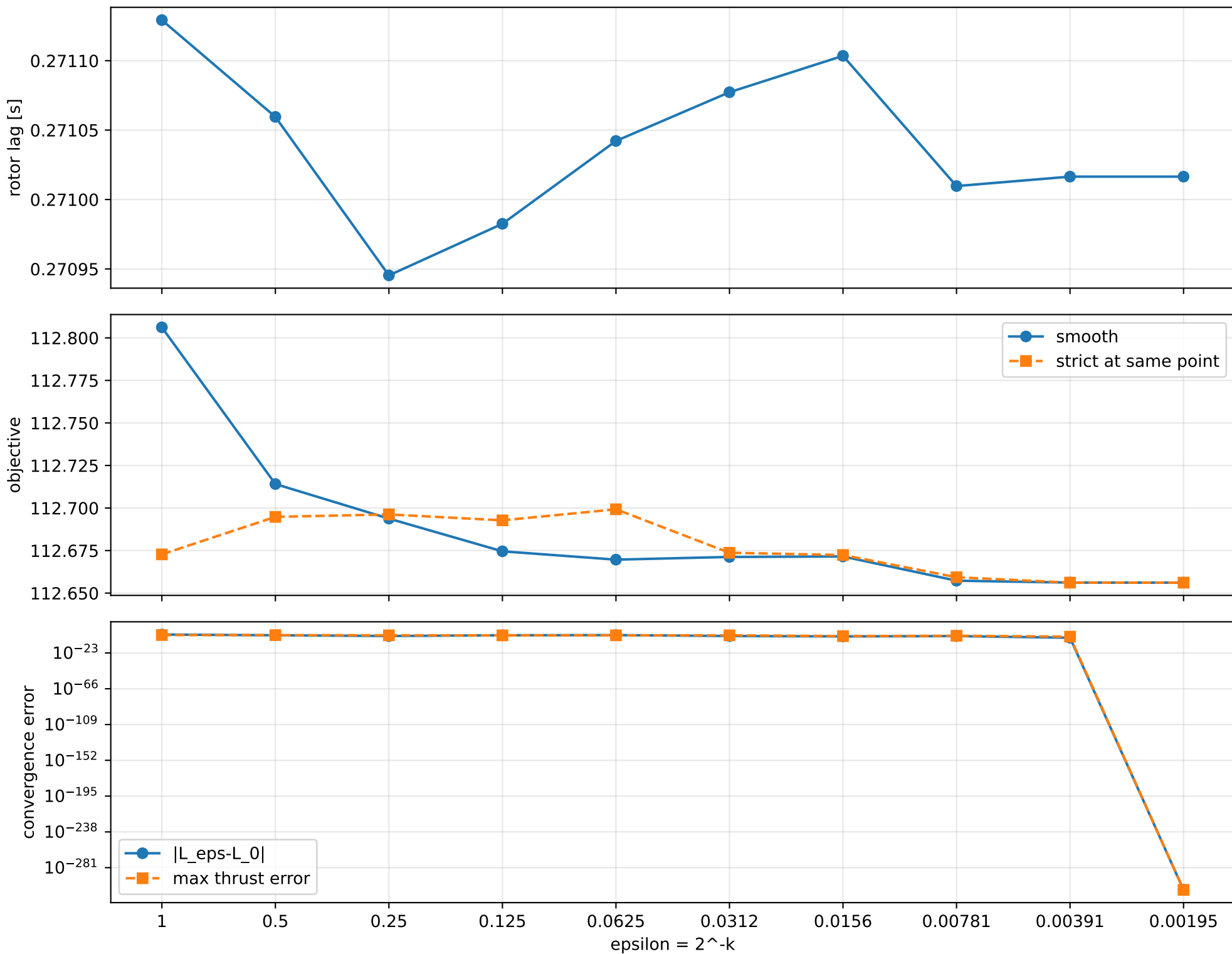
gyro [rad/s]



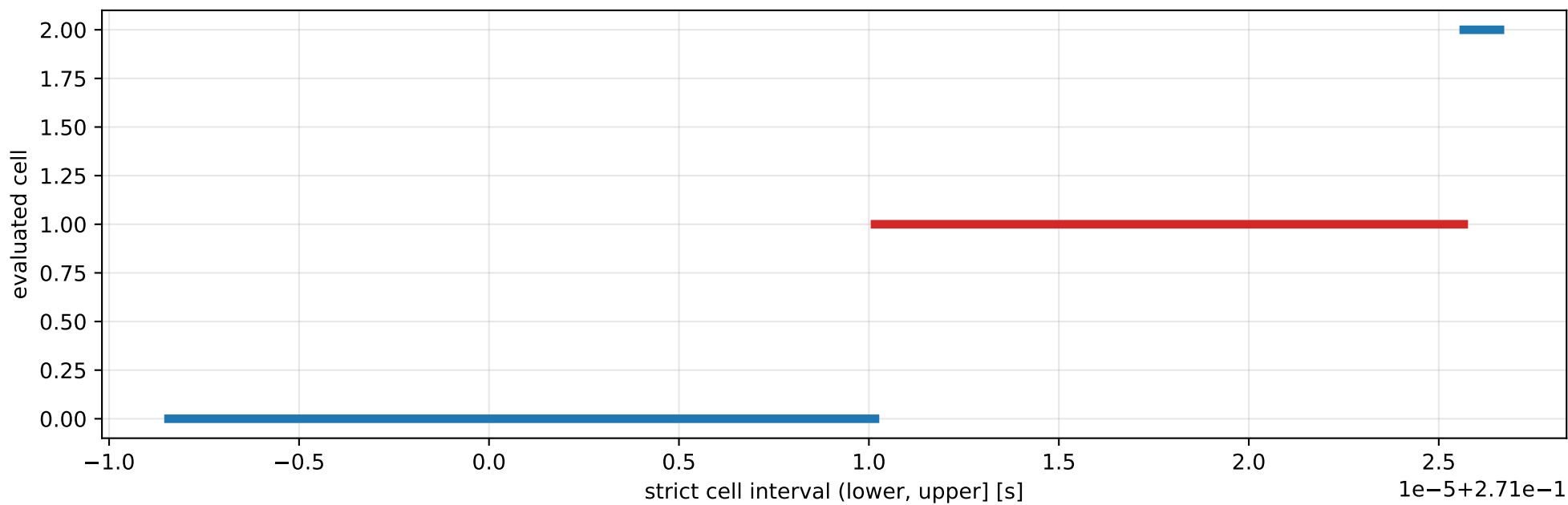
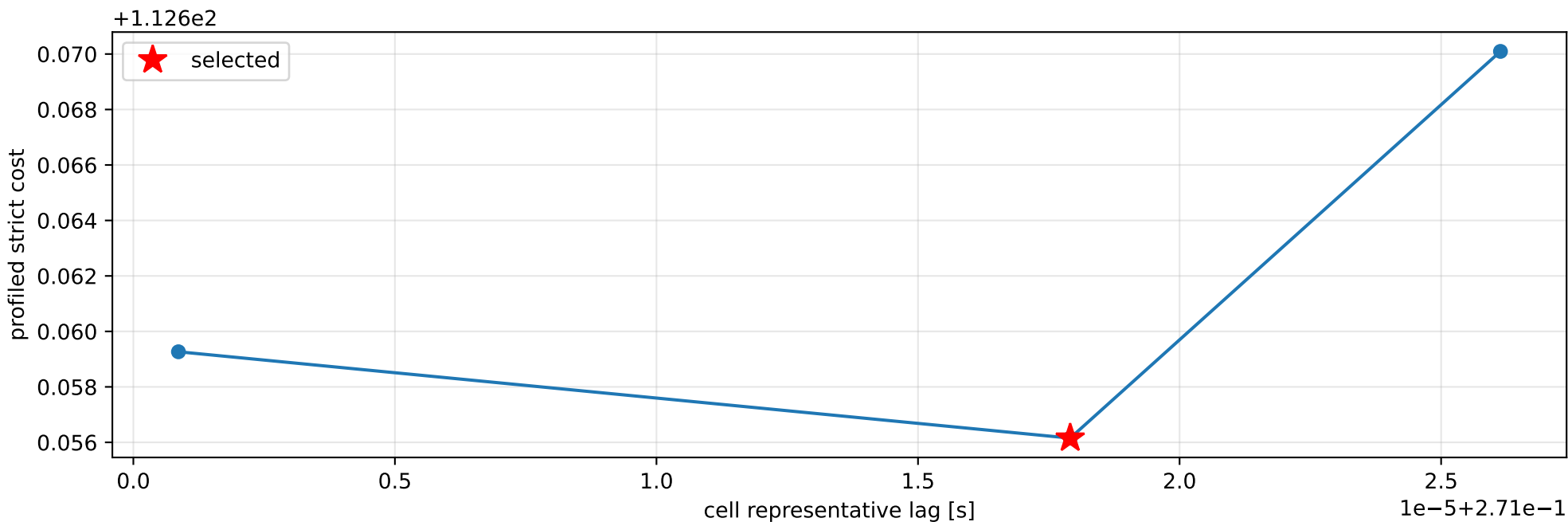
specific force [m/s²]



so3_naive_rotation_vector_derivatives: smooth-to-strict continuation



so3_naive_rotation_vector_derivatives: exact strict-ZOH lag cells



selected (0.27101016, 0.271025658] s; final smooth 0.271016482 s; data boundary=False

so3_naive_rotation_vector_derivatives: parameter estimate

Case: so3_naive_rotation_vector_derivatives

status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 8.36667904

inertia [kg m²]:

```
[[ 0.61720467  0.01526055 -0.0030274 ]
 [ 0.01526055  0.29120115 -0.09126026]
 [-0.0030274  -0.09126026  0.88059036]]
```

CoG body [m]: [-0.02641846 0.01135696 0.00938901]

force effectiveness: [3.19426896 2.58508329 2.85264802 3.42544269]

scale-free J/m [m²]:

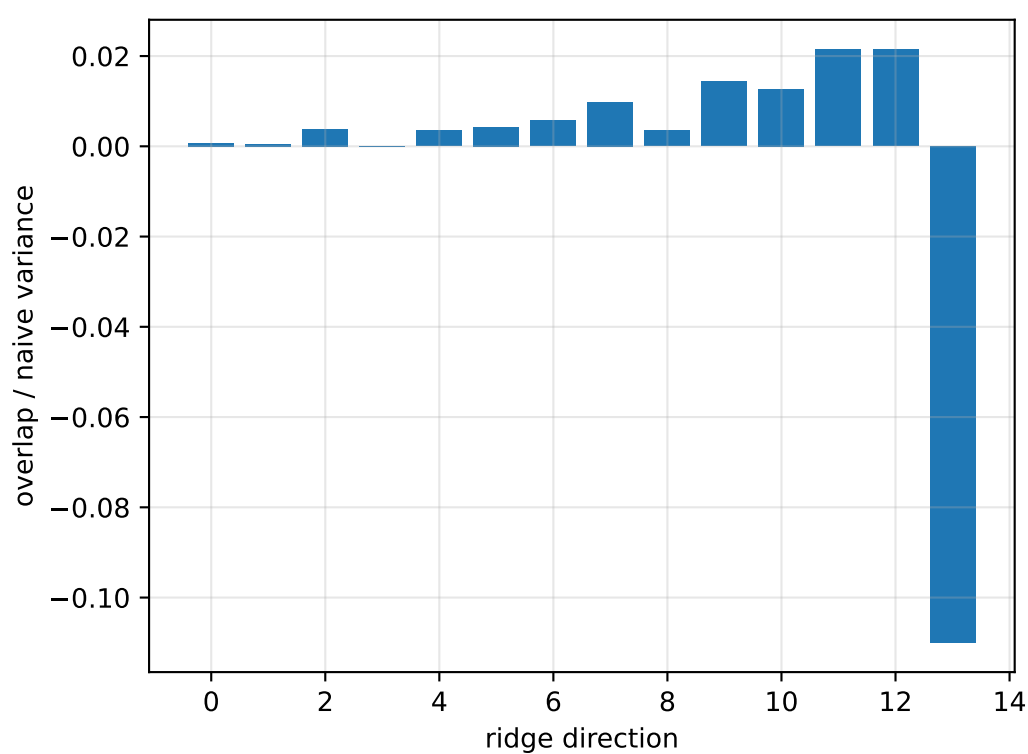
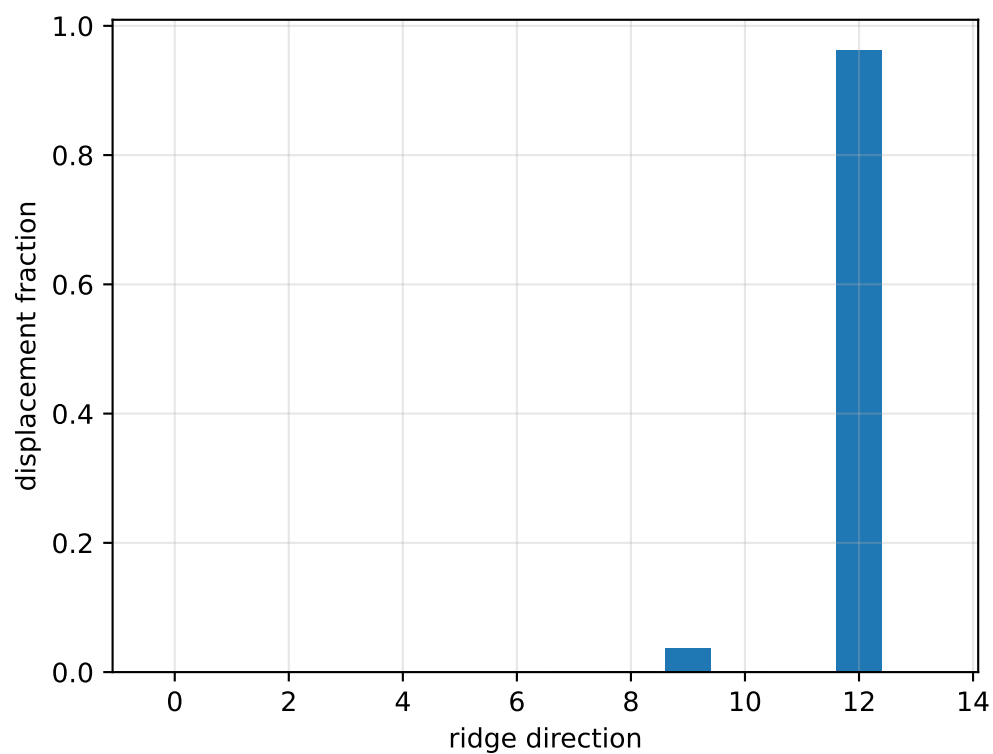
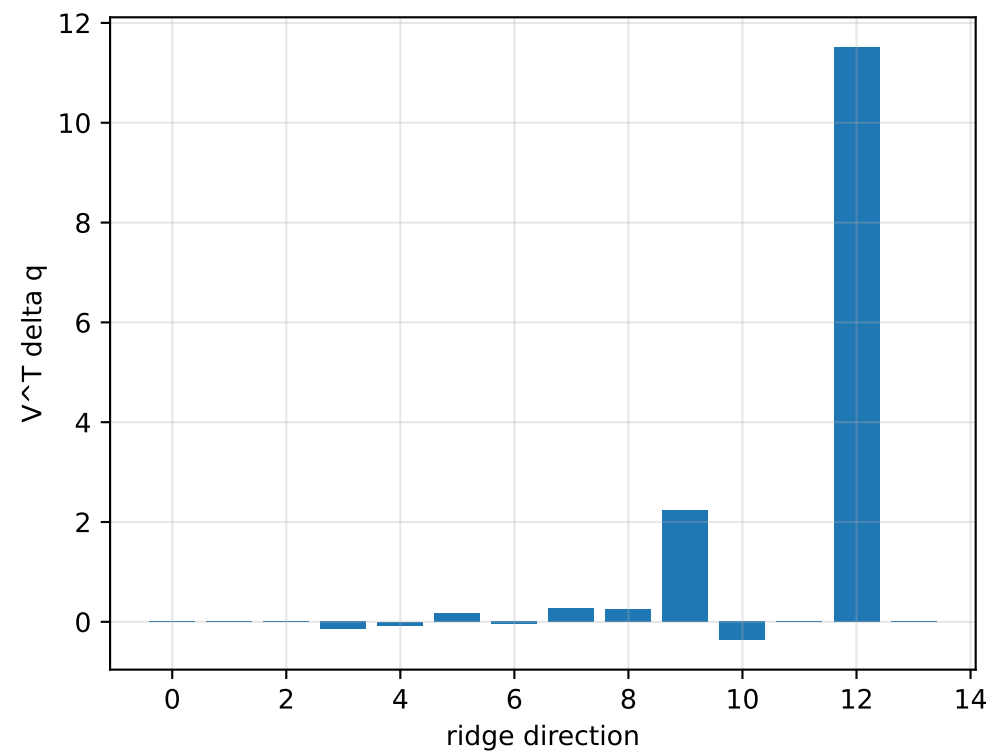
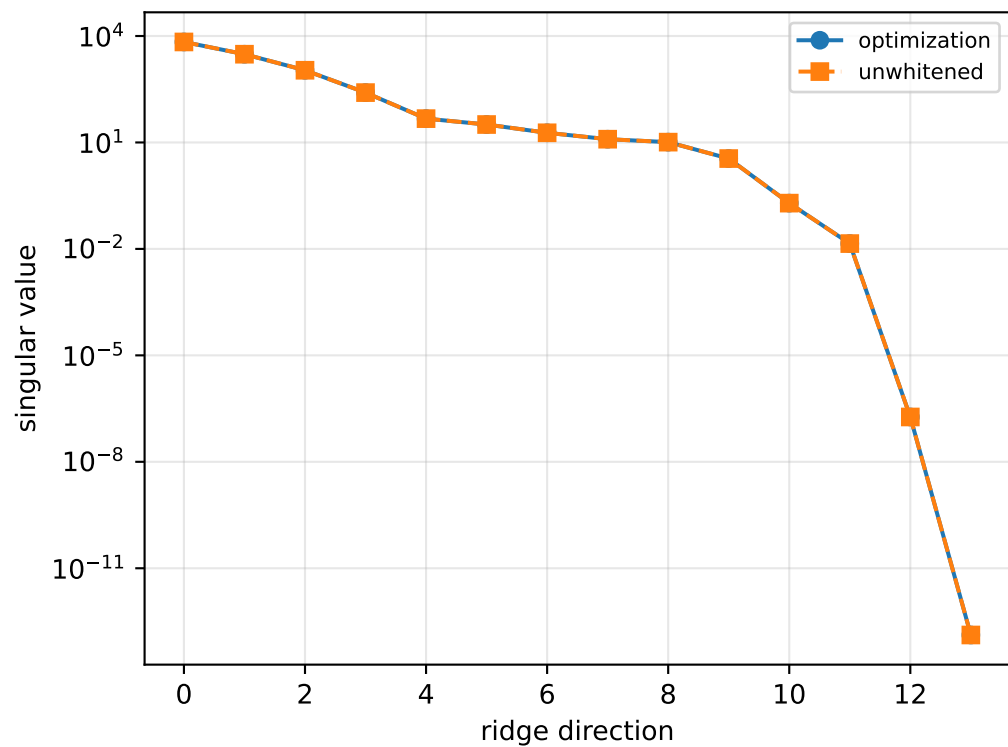
```
[[ 0.07376937  0.00182397 -0.00036184]
 [ 0.00182397  0.03480487 -0.01090759]
 [-0.00036184 -0.01090759  0.10524969]]
```

scale-free f/m: [0.38178457 0.30897364 0.34095344 0.40941486]

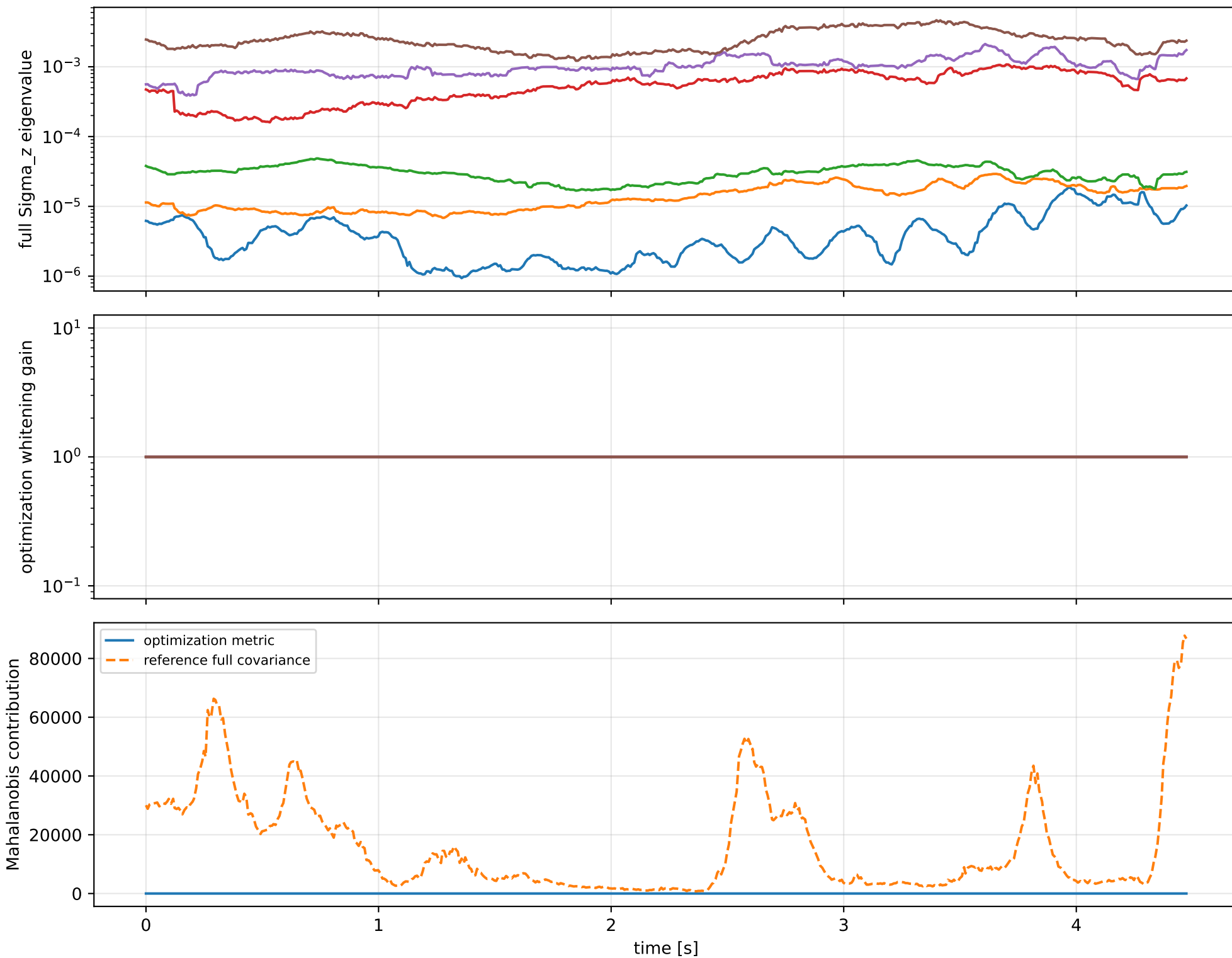
rotor lag cell: (0.27101016, 0.271025658] s

representative: 0.271017909 s

so3_naive_rotation_vector_derivatives: ridge spectrum (rank 13, nullity 1, ||v||=1.57e-13)



so3_naive_rotation_vector_derivatives: optimization vs reference covariance



so3_naive_rotation_vector_derivatives: wrench / closure (force max $2.32\text{e-}14$, torque max $3.47\text{e-}16$)

